ARJUNA JEE 2027

Lecture -09 | Mathematics

PERMUTATIONS AND COMBINATIONS

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Topics *to be covered*

A **Division & Distribution**

B **Question Practice**

C

D



H.W. Discussion

Question [JEE Mains-2026]

[Ans. : 126]

Two players A and B play a series of games of badminton, The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is _____.

Question [JEE Mains-2026]

[Ans. : 660]

Let ABC be a triangle. Consider four points p_1, p_2, p_3, p_4 on the side AB, five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC. None of these points is a vertex of the triangle ABC. Then the total number of pentagons, that can be formed by taking all the vertices from the points $p_1, p_2, \ldots, p_{13}$, is



Question

Find the number of arrangements of letters of the word **EQUATION** such that vowels occur in alphabetical order.



Question

When repetition is allowed find the number of 7 digits numbers 0 to 9

(i) such that sum of digits is even

(ii) which are palindrome.



Question [JEE Mains-2020]

The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word EXAMINATION is _____.



Question [JEE Mains-2026]

[Ans. : (D)]

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is

A 2670

B 2840

C 2920

D 3600





DIVISOR PROBLEMS

$N = 2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma} \cdot 7^{\delta} \ldots\ldots$

(I) Total number of divisors = $(\alpha + 1).(\beta + 1).(\gamma + 1).(\delta + 1)\ldots\ldots$

(II) Total number of proper divisors = $(\alpha + 1).(\beta + 1).(\gamma + 1).(\delta + 1)\ldots\ldots - 2$

(III) Sum of all the divisors = $(2^0+2^1+2^2+\ldots+2^{\alpha}).(3^0+3^1+\ldots+3^{\beta}).(5^0+5^1+\ldots+5^{\gamma})\ldots.$

(IV) Number of ways in which 'N' can be resolved as product of two factors

$= \left(\frac{\text{total divisors}}{2}\right)$, 'N' $\equiv$ "is not a perfect square"

$= \left(\frac{\text{total divisors} + 1}{2}\right)$, 'N' $\equiv$ "is a perfect square" (Hint: $\alpha, \beta, \gamma, \delta\ldots \rightarrow$ all are even)

(V) Number of ways in which 'N' can be resolved in product of two co-prime factors = 2^{k-1}, (where k = total no. of prime factors)

Bumper Question

If $N = 2^4.3^3.5^2.6^3$ $\Rightarrow 6^3 = 2^3.3^3$ $\Rightarrow N = 2^7.3^6.5^2$

1. (i) Total number of divisors $= (7+1).(6+1).(2+1)$ ✓
   (ii) total number of divisors which are perfect square $= (3+1).(3+1).(1+1)$
   (iii) Total number of divisors which are perfect cube. $= (2+1).(2+1) = 9$
2. (i) Total number of divisors divisible by 3.
   (ii) Total number of divisors divisible by 15
   (iii) Total number of divisors divisible by 12

1) iii) $N = 2^1 \boxed{8^2\ 27^2}.\ 5^2$

1) ii) $N = 2^1 \boxed{4^3.9^3\ 25^1}$

atleast one '3' is selected.

2(i) $(7+1).6.(2+1)$

atleast one 3 & one 5 is selected.

2(ii) $\boxed{(7+1).6.2}$

$\Rightarrow \boxed{N = 2^7 \cdot 3^6 \cdot 5^2}$

2) (iii)

$N = 2^1 \cdot \underline{\underline{4^3}} \cdot 3^6 \cdot 5^2$

$\boxed{(1+1) \cdot 3 \cdot 6 \cdot (2+1) = 108}$ ✓

OR

$3 \cdot 6 \cdot (2+1) = 54$ ✗

$\# \underline{12} = 2^2 \cdot 3^1$

$N = 2^7 \cdot 3^6 \cdot 5^2$

3. Sum of all divisors $= (2^0+2^1+\ldots+2^7)\cdot(3^0+3^1+\ldots+3^6)\cdot(5^0+5^1+5^2)$

(i) Sum of all even divisors $= (2^1+2^2+\ldots+2^7)\cdot(3^0+3^1+\ldots+3^6)\cdot(5^0+5^1+5^2)$

(ii) Sum of all proper odd divisors $= 2^0\cdot(3^0+3^1+\ldots+3^6)(5^0+5^1+5^2) - (1)$

(iii) Sum of all divisors divisible by 10 $= (2^1+2^2+\ldots+2^7)\cdot(3^0+3^1+\ldots+3^6)\cdot(5^1+5^2)$

$2^1\cdot5^1$

4. Number of ways in which 'N' can be resolved as product of two number's

not a P.S.

$$\frac{\text{Total div.}}{2}$$

$N = 2^7 \cdot 3^6 \cdot 5^2$

5. Number of ways in which 'N' can be resolved as product of two co-prime number's

$k = 3$

$2^{3-1} = 4$

4 ways

$\{1, 2^7 \cdot 3^6 5^2\}$, $\{2^7, 3^6 5^2\}$, $\{3^6, 2^7 5^2\}$, $\{5^2, 2^7 3^6\}$

6. (i) Number of divisors of the form $(4n)$ =

(ii) Number of divisors of the form $(4n+2)$ = $1 \cdot (6+1) \cdot (2+1) = 21$

div. by 4: $N = 2^1 \cdot 4^3 \cdot 3^6 5^2$ (2^7)

$\text{Inf} = (1+1) \cdot 3 \cdot (6+1) \cdot (2+1)$

$4n+2 = 2^1 \cdot (2n+1)$

1 mein ek hi '2' present hai.

$D_1 = 2 \cdot 3^2 \cdot 5^1 = 90 = 4 \times 22 + 2$

$D_2 = 2^1 \cdot 3^1 \cdot 5^2 = 150$

$D_3 = 2^1 \cdot 5^1 = 10$

6. (iii) Number of divisors of the form $(4n+1)$ =

$N = 2^7 \cdot 3^6 5^2$

2^0 2^1 2^2 2^3 2^4 ... 2^7

$4n+1$ $4n+2$ $4n$ $4n$ $4n$ $4n$

3^0 3^1 3^2 3^3 3^4 3^5 3^6

$4n+1$ $4n+3$ $4n+1$ $4n+3$ $4n+1$ $4n+3$ $4n+1$

5^0 5^1 5^2

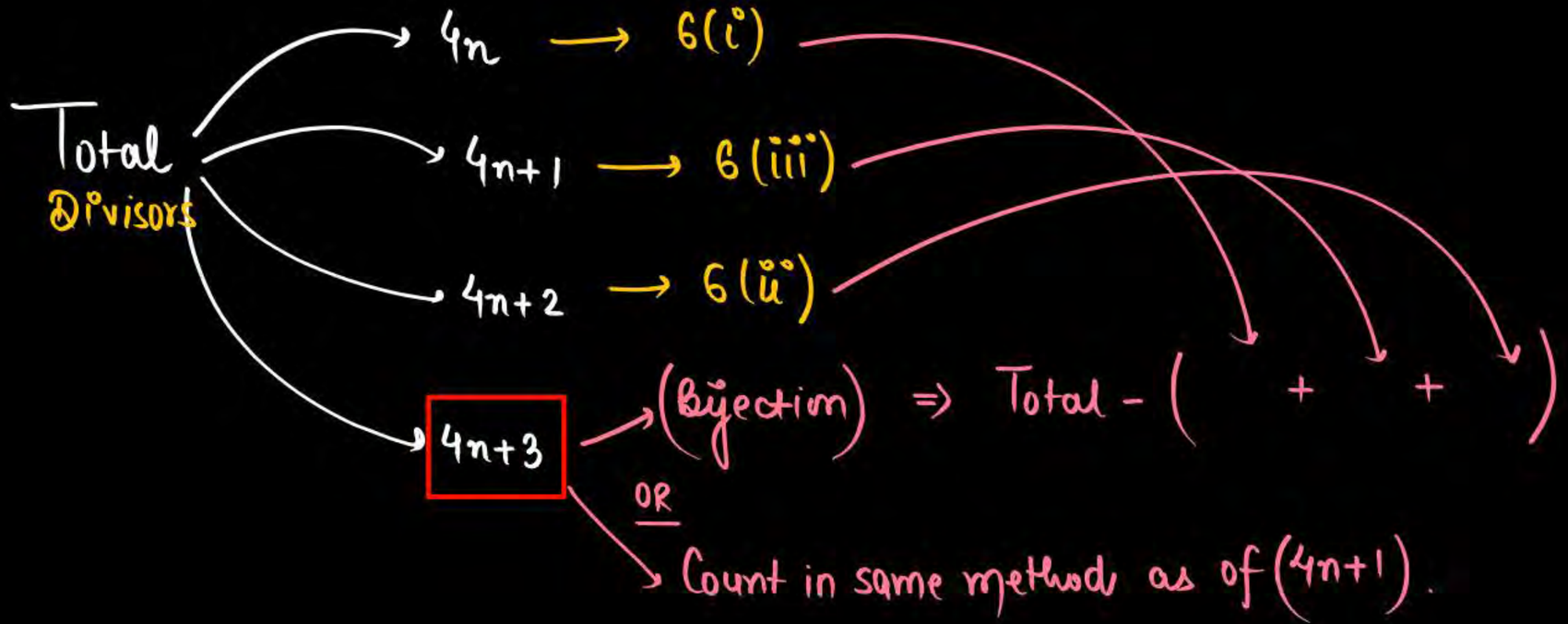
$4n+1$ $4n+1$ $4n+1$

$(4n+1)$ = '2' ki kuch power × 3 × 5

$4n+1$ $4n+1$ $4n+1$

$1 \times 4 \times 3 = 12$ fnf.

$4n+1 \cdot 4n+3 \cdot 4n+3$ — n.p.


Total
Divisors
4n → 6(i°)
4n+1 → 6(iii°)
4n+2 → 6(ii°)
4n+3
(Bijection) ⇒ Total - (+ +)
OR
Count in same method as of (4n+1).

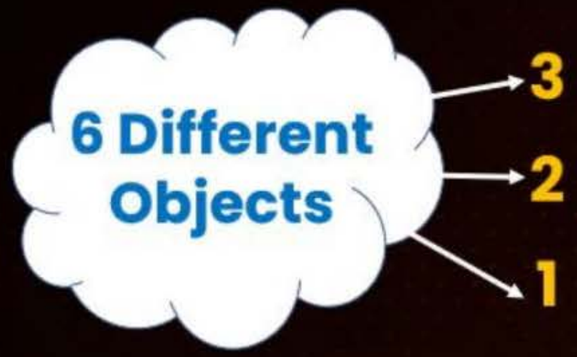
Question [JEE Mains-2023]

Let n be a non-negative integer. Then the number of divisors of the form $4n + 1$ of the number $(10)^{10} \cdot (11)^{11} \cdot (13)^{13}$ is equal to _____.



DIVISION INTO GROUPS

Case-I: When group size is different



$$\Rightarrow \text{No. of ways} = \frac{6!}{3!\,2!\,1!}$$



$$\Rightarrow \text{No. of ways} = \frac{14!}{5!\,4!\,3!\,2!}$$

DIVISION INTO GROUPS

Case-II: When group size is same

6 Different Objects → 2, 2, 2

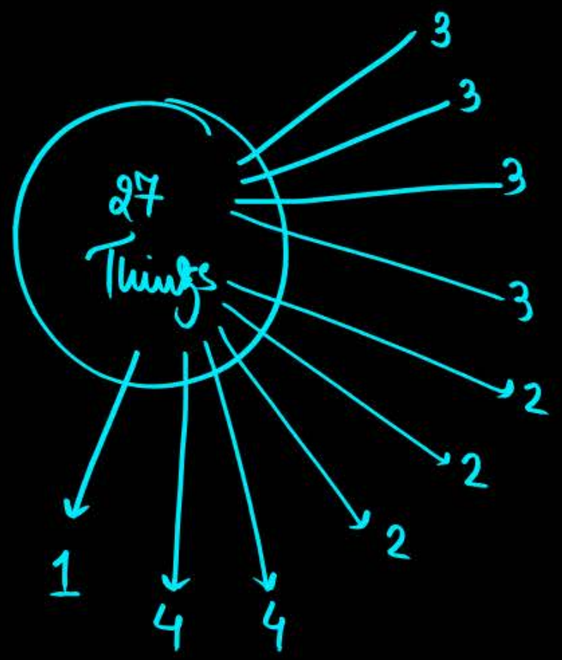
$$\Rightarrow \text{No. of ways} = \frac{6!}{2!\,2!\,2!\,3!}$$

7 Different Objects → 2, 2, 1, 1, 1

$$\Rightarrow \text{No. of ways} = \frac{7!}{2!\,2!\,1!\,1!\,1!\,2!\,3!}$$

10 Different Objects → 3, 3, 2, 2

$$\Rightarrow \text{No. of ways} = \frac{10!}{3!\,3!\,2!\,2!\,2!\,2!}$$



$$\frac{27!}{(3!3!3!3!\cdot 4!)(2!2!2!\cdot 3!)(4!4!2!)\,1!}$$

Example

Number of ways in which 52 playing cards equally divided into:

(i) Four heaps

$$\frac{52!}{13!\,13!\,13!\,13!\times 4!}$$

(ii) Four Persons

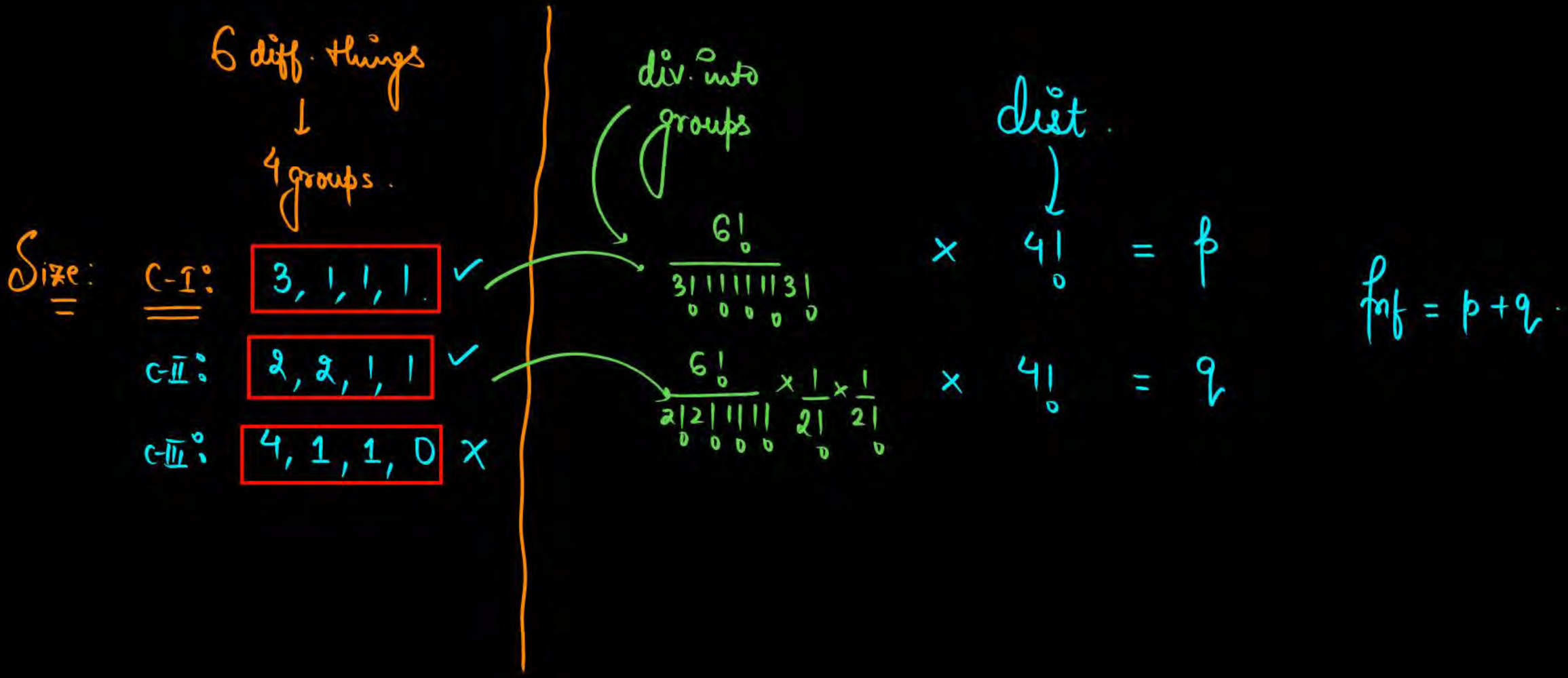
divide into groups.

distribution

$$\frac{52!}{13!\,13!\,13!\,13!\,4!}\times 4!$$

Example

In how many ways six different books can be distributed between four persons, so that each person gets atleast one book.





$$\underset{B_1}{\frac{4}{}}\ \underset{B_2}{\frac{4}{}}\ \underset{B_3}{\frac{4}{}}\ \underset{B_4}{\frac{4}{}}\ \underset{B_5}{\frac{4}{}}\ \underset{B_6}{\frac{4}{}} \rightarrow 4^6 \text{ fnf.}$$

Example

Number of ways of distributing seven different chocolates to four children such that each gets atleast one chocolate.



7 diff. things
↓
" 4 groups "

Question [JEE (Adv.)-2020, Paper-2]

In a hotel, four rooms are available. Six persons are to be accommodated in these four rooms in such a way that each of these rooms contains at least one person and at most two persons. Then the number of all possible ways in which this can be done is_____.



6 persons

2, 2, 1, 1 ✓

3, 1, 1, 1 ✗

2, 2, 2, 0 ✗

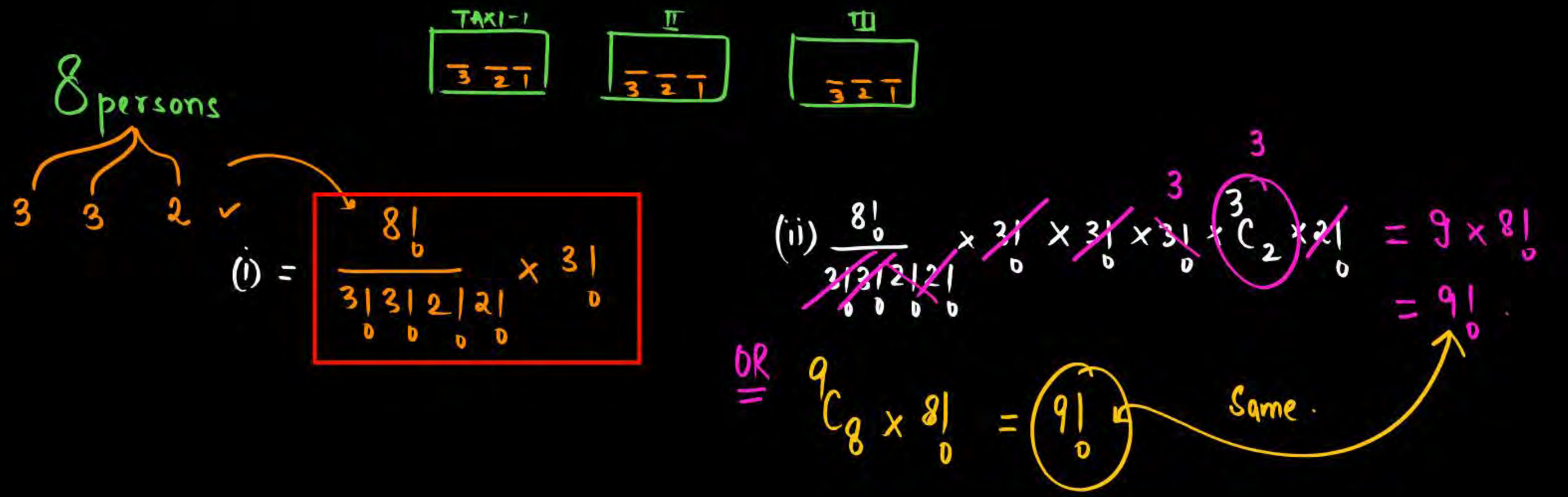
$$\frac{6!_0}{2!2!1!1! \times 2!2!} \times 4!_0 = \text{Ans.}$$

Acidic Question

Number of ways in which 8 persons can be seated in three different taxies each having 3 seats for passengers and duly numbered is

(a) If internal arrangement of persons inside the taxi is immaterial.

(b) If internal arrangement also matters.



DO IT BY YOURSELF (DIBY)

DIBY-46

Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from all 4 different sides as vertices and β be the number of quadrilaterals having these points from all 4 different sides as vertices. Then $(\beta - \alpha)$ is equal to [JEE Mains-2021]

(A) 795 (B) 1173 (C) 1890 (D) 717

[Ans. : (D)]

DIBY-47

There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is _____. [JEE Mains-2024]

[Ans. : 5626]

DIBY-48

The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is [JEE Mains-2023]

(A) 89 (B) 84 (C) 86 (D) 79

[Ans. : (A)]

DIBY-49

All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is [JEE Mains-2023]

(A) 327 (B) 328 (C) 324 (D) 326

[Ans. : (A)]

DIBY-50

All the letters of the word PUBLIC are written in all possible orders and these words are written as in a dictionary with serial numbers. Then the serial number of the word PUBLIC is [JEE Mains-2023]

(A) 576 (B) 578 (C) 580 (D) 582

[Ans. : (D)]

DIBY-51

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is [JEE Mains-2022]

[Ans. : 1492]

DIBY-52

All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of the word "GTWENTY" is [JEE Mains-2024]

[Ans. : 553]

DIBY-53

60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50th word is:

[JEE Mains-2024]

(A) JBBOH (B) OBBJH (C) OBBHJ (D) HBBJO

[Ans. : (B)]

DIBY-54

If all the words with or without meaning made using all the letters of the word "NAGPUR" are arranged as in a dictionary, then the word at 315th position in this

[JEE Mains-2024]

(A) NRAGUP (B) NRAPUG (C) NRAPGU (D) NRAGPU

[Ans. : (C)]

DIBY-55

Find the total number of ways of selecting five letters from the work 'INDEPENDENT'.

[Ans. : 72]

DIBY-56

The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to [JEE Mains-2024]

(A) 179 (B) 177 (C) 181 (D) 175

[Ans. : (A)]

DIBY-57

The total number of words (with or without meaning) that can be formed out of the letters of the word 'DISTRIBUTION' taken four at a time, is equal to

[JEE Mains-2024]

[Ans. : 3734]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

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